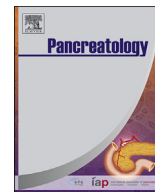




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The role of hypertriglyceridemia for acute kidney injury in the course of acute pancreatitis and an animal model

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ABSTRACT

Objective: The aim of this study was to investigate the role of hypertriglyceridemia for acute kidney injury (AKI) in the course of acute pancreatitis.

Methods: Patients with acute pancreatitis were retrospectively divided into four groups according to admission triglyceride: normal group, mild HTG group, moderate HTG group and severe HTG group. Clinical characteristics were compared among these groups. Wild type (WT) mice and Human ApoC III transgenic (ApoCIII^{Tg}) mice were used in the next animal experiments. Severe acute pancreatitis (SAP) model was established by retrograde injection of 0.5% sodium taurocholate (0.1 ml/100 g) from duodenum to pancreatic duct. Histological scores, serum amylase, creatinine, urea nitrogen were compared between WT mice and ApoCIII^{Tg} mice.

Results: Two hundred and sixty-two patients were classified into 4 groups: normal TG (104, 39.7%), mild HTG (72, 27.5%), moderate HTG (47, 17.9%), and severe HTG (39, 14.9%) groups. The proportions of AKI were 13.5% (14/104, normal), 13.9% (10/72, mild), 21.3% (10/47, moderate), and 38.5% (15/39, severe), respectively. After establishing SAP model, the levels of serum amylase ($P < 0.05$) and pancreatic histological score ($P < 0.05$) of ApoCIII-SAP-9h group were significantly higher than that of WT-SAP-9h group, respectively. ApoCIII-SAP-9h group had significantly higher levels of serum creatinine ($P < 0.001$), urea nitrogen ($P < 0.001$), and kidney histological score ($P < 0.05$) than that of WT-SAP-9h group, respectively.

Conclusions: Mild HTG has little adverse impact on disease severity of acute pancreatitis; severe HTG can aggravate kidney injury in the course of acute pancreatitis. ApoCIII-SAP mice have more serious pancreatic damage and kidney injury than WT-SAP mice.

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Introduction

Hypertriglyceridemia (HTG) is an accompanying condition in acute pancreatitis (AP) accounting from 2% to 26% [1]. It has been regarded as one of the most important issues, relating to the incidence, disease course and recurrence of AP [2,3]. Although lots of studies focus on HTG as an etiology of AP, the relationship of HTG and disease severity is not well established.

A recent systemic review indicated studies comparing the severity of hypertriglyceridemic AP with disease caused by other

etiologies are heterogeneous and scarce [1]. In the study by Balachandra et al. [4], HTG had no significant correlation with the clinical severity of AP. Fortson et al. [5] reported that the prevalence of local complications in patients with HTG was similar to AP from other causes. However, Goyal et al. [6] found hypertriglyceridemic AP had greater severity of disease and less favorable outcomes than alcoholic pancreatitis. Elevated serum TG is independently and proportionally correlated with persistent organ failure regardless of etiology of AP [7]. Several other studies also showed that patients with HTG tended to have more severe disease course of AP [8–12]. An important reason of the heterogeneity is that few studies classify the level of serum TG more particular.

Acute kidney injury (AKI) takes a proportion of 20–25% of all AP population, and the total mortality of AP patients with AKI is about 25% [13,14]. Our previous study indicated HTG is an independent

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risk factor for AKI in the early phase of AP [15]. We would further classify HTG into more detailed groups, and investigate the relationship of HTG and AKI. Havel et al. [3] supposed that free fatty acids (FFAs) hydrolyzed by pancreatic lipase from triglyceride (TG) is a potential cause of hypertriglyceridemic pancreatitis. Kimura et al. [16] also found pancreatic damage initiated by different pathogenetic pathways can be worsened by TG in animal models. We proposed that a similar pathophysiology may occur in hypertriglyceridemic pancreatitis related AKI.

The experimental studies almost unanimously confirm that HTG can aggravate AP in animal models. High-fat diet feeding is the most common method to simulate HTG condition. However, the serum level of TG can be only slightly higher than normal range in the above condition, and accordingly, it does not much conform to the clinical characteristics of hypertriglyceridemic AP. Human ApoC III transgenic (ApoCIII^{Tg}) mice were well adopted in the study for hypertriglyceridemic AP as an ideal animal model of primary severe HTG [17]. We would investigate the role of HTG for AP-related AKI from clinical study and animal experiments.

Materials and methods

Clinical study

Patients

From July 2009 to July 2014, patients with a diagnosis of AP admitted to the department of general surgery of Jinling Hospital were retrospectively included and analyzed. AP was diagnosed according to the serum amylase (Amy) level, combined with clinical presentations and image findings from computed tomography. The severity of AP were classified according to the revised Atlanta criteria [18]. Inclusion criteria were AP patients admitted to our hospital within 48 h of disease onset. Patients with recurrent AP and those lacking necessary clinical and biochemical data were excluded.

Data collection and clinical manifestations

Demographic parameters included age, gender, body mass index (BMI), history of smoking and alcohol abuse. Chronic comorbidities included hypertension, diabetes, HTG, biliary diseases, cardiovascular diseases, respiratory diseases and chronic renal diseases. Demographic parameters and chronic comorbidities were collected and analyzed. BMI was calculated as weight in kilograms divided by height in meters squared (kg/m^2). Data were recorded on standard admission proformas and supplemented by retrospective chart review from the hospital laboratory database. Organ functions were evaluated on three organ systems and organ failure was defined as acute respiratory distress syndrome (ARDS) ($\text{PaO}_2/\text{FiO}_2 \leq 300 \text{ mmHg}$), AKI (creatinine $\geq 2.0 \text{ mg/dl}$) and shock (need for vasoactive agent). The incidence of infected pancreatic necrosis, requirement for surgical intervention, length of intensive care unit (ICU) and hospital stay as well as mortality of the AP patients were also recorded.

Gratation of serum TG

Patients were divided into four groups according to admission serum TG: normal ($<150 \text{ mg/dL}$), mild HTG ($150\text{--}500 \text{ mg/dL}$), moderate HTG ($500\text{--}1000 \text{ mg/dL}$), and severe HTG ($>1000 \text{ mg/dL}$) groups.

Animal experiments

Animals and groups

Eight-week-old male wild type (WT) ICR mice and male ApoC-III^{Tg} mice in ICR background were obtained from Institute of

Cardiovascular Sciences, School of Basic Medical Sciences, Peking University Health Science Center. In ApoCIII^{Tg} group, only mice with serum TG more than 500 mg/dL were kept for next experiments. All mice were maintained in a temperature-controlled room ($22 \pm 1^\circ\text{C}$) with a 12 h light/dark cycle for adaptation of 2 weeks. Both 30 WT mice and 30 ApoCIII^{Tg} mice were randomly divided into 5 groups: Control group, SAP-3h group, SAP-6h group, Sham-9h group, and SAP-9h group. Each group had 6 mice. All the animals were deprived of food and drinking water for 3 h before operation.

Model of SAP

Mice in control group were sacrificed under normal condition. SAP model was established by retrograde injection of 0.5% sodium taurocholate ($0.1 \text{ ml}/100 \text{ g}$ weight) from duodenum to pancreatic duct as method of Perides et al. [19]. Sham group was established using normal saline by the same way. All mice underwent surgery under 2% pentobarbital sodium ($0.1 \text{ ml}/100 \text{ g}$) intraperitoneal injection anesthesia and sterile conditions.

Tissue samples

Pancreas and kidneys were quickly collected into liquid nitrogen, and stored at -80°C for analysis. A tail of the head of the pancreas and the extracted kidney were fixed in 10% buffered formalin, and embedded in paraffin for morphological evaluation by H&E staining. Sections were slid to $5 \mu\text{m}$ -thick pieces and coded.

Pancreas morphological evaluation

The degree of pancreatic injury by light microscopy was based on severity of edema, inflammatory cell infiltration, and cell death. A pathologist blinded to the experiment evaluated the pathologic scores according to Schmidt's criteria [20].

Kidney morphological evaluation

The coded kidney specimens were stained with hematoxylin and eosin and examined in blinded fashion. Histological changes were evaluated by quantitative measurement of tubulointerstitial injury, which was assessed by counting the number of necrotic and apoptotic cells, loss of tubular brush border, tubular dilatation, cast formation, and neutrophil infiltration. The scoring was 0 = none; 1 = 0–10%; 2 = 11–25%; 3 = 26–45%; 4 = 46–75%; and 5 = 76–100% [21].

Assessment of blood chemistries

Blood was sampled from mouse eyes when mice were sacrificed, and was centrifuged at $15,000 \text{ rpm}$ for 5 min in a microfuge. Total plasma TG was measured by the L type TG H kit (Jiancheng, Nanjing). High concentration of serum TG could interfere the evaluation of other biomarkers. Density-gradient ultracentrifugation was used to separate plasma lipoprotein for further detecting. The plasma samples were centrifuged for 18 h at 5°C at $40,000 \text{ rpm}$ in a Ti-50 fixed-angle rotor (Beckman Instruments, Geneva, Switzerland). After ultracentrifugation, serum Amy, creatinine (Scr) and blood urea nitrogen (BUN) were detected to evaluated pancreas damage and renal functions. Serum Amy level was determined using a colorimetric assay kit (Jiancheng, Nanjing). Scr and BUN concentrations were determined by the Scr and BUN Companion kit (Exocell).

Statistical analysis

Statistical analysis was performed using SPSS version 12 statistical software. Continuous variables were expressed as mean $\pm$ SEM or as medians with interquartile rang, and are

analyzed with two tailed *t*-test for normal distributions and the Mann-Whitney *U* test for non-normal distributions. Categorical data were analyzed by using χ^2 test or Fisher's exact test as appropriate. Multivariate logistic regression analysis (enter method) was performed to estimate the odds ratio of HTG as a risk factor for moderate and severe AP. A *P* value < 0.05 was considered to be of statistical significance.

Results

Clinical study

HTG has a close relationship with disease severity of AP

There were 262 patients with AP, and they were divided into mild AP (MAP) (143, 54.5%), moderate AP (67, 25.6%), and SAP (52, 19.8%) according to the revised Atlanta criteria. The average age of the patients was 45.9 ± 14.6 years, and there were 160 males (61.1%). According to admission serum TG level, all patients were classified into four groups: normal TG (104, 39.7%), mild HTG (72, 27.5%), moderate HTG (47, 17.9%), and severe HTG (39, 14.9%) groups. Basic characteristics of the patients in these four groups had no statistical differences (Table 1). From normal TG to severe HTG, the proportions of the patients with alcohol abuse were 21.0%, 23.6%, 29.8%, and 35.9%, respectively. Pearson analysis revealed positive correlation between alcohol abuse and serum TG levels (Pearson correlation coefficient value was 0.965, *P* = 0.035). Multivariate logistic regression analysis involving alcohol abuse and HTG showed that HTG was an independent risk factor for moderate AP combined with SAP (odds ratio (OR) = 2.432; 95% confidence interval (CI) 1.358–4.410; *P* = 0.001), while the role of alcohol abuse (OR = 1.372; 95% CI 0.794–2.653; *P* = 0.217) was relatively weak. The incidences of ARDS increased from normal group to severe HTG group. About complications of shock and IPN,

no statistical differences were shown among patients between each two groups (Table 2). However, patients in severe HTG group had statistically higher proportions of surgical intervention, needing longer length of ICU stay and hospital stay than normal and mild HTG groups. The mortality rate of patients in severe HTG group was 12.8%, much higher than other groups.

Only severe HTG is associated with significantly higher proportion of AKI

The proportion of AKI in normal group was 13.5% (14/104). In mild HTG group, the proportion of AKI was 13.9% (10/72), which was similar to the result of normal group. The proportion increased to 21.3% (10/47) in moderate HTG group, and continued on its up trend to 38.5% (15/39) in severe HTG group, which was significantly higher than that of normal and mild HTG groups.

Animal experiments

The mean level of serum TG of ApoCIII-control group was significantly higher than that of WT-control group (*P* < 0.001) (Fig. 1A).

Comparison of serum Amy between Sham-9h group and SAP-9h group

In order to exclude the effects of operation, we compared serum Amy between sham-9h group and SAP-9h group. For both ApoCIII^{tg} and WT mice, the levels of serum Amy (Fig. 2) of SAP-9h group were significantly higher than that of sham-9h group (*P* < 0.001).

ApoCIII-SAP groups had more serious pancreatic damage than WT-SAP groups

Serum levels of Amy (Fig. 3A) were compared between ApoCIII-SAP groups and WT-SAP groups. The levels of serum Amy of

Table 1

Baseline characters of patients with different levels of admission serum triglyceride.

	Normal, n = 104	Mild HTG, n = 72	Moderate HTG, n = 47	Severe HTG, n = 39
Gender	65 (62.5)	43 (59.7)	28 (59.6)	24 (61.9)
Age	46.4 ± 14.4	46.0 ± 15.0	45.9 ± 15.9	46.2 ± 13.2
Hypertension	24 (22.9)	17 (23.6)	9 (19.1)	10 (25.6)
BMI	25.2 ± 2.8	25.5 ± 3.5	25.8 ± 3.7	25.6 ± 3.2
Diabetes	13 (12.5)	10 (13.8)	5 (10.6)	6 (15.4)
Cardiovascular diseases	4 (3.8)	3 (4.2)	2 (4.3)	2 (5.1)
Respiratory diseases	3 (2.9)	3 (4.2)	1 (2.1)	1 (2.6)
Renal diseases	6 (5.8)	7 (9.7)	5 (10.6)	3 (7.7)
Etiologies				
Biliary diseases	41 (39.0)	27 (37.5)	21 (44.7)	11 (28.2)
Alcohol abuse	22 (21.0)	17 (23.6)	14 (29.8)	14 (35.9)
Others	41 (39.0)	28 (28.9)	12 (25.5)	14 (35.9)

**P* < 0.05 compared with normal group; #*P* < 0.05 compared with mild HTG group; †*P* < 0.05 compared with moderate HTG group. BMI, body mass index; TG, triglyceride.

Table 2

Clinical manifestations of patients with different levels of admission serum triglyceride.

	Normal, n = 104	Mild HTG, n = 72	Moderate HTG, n = 47	Severe HTG, n = 39
ARDS	23 (21.9)	13 (18.1)	14 (29.8)	12 (30.7)
AKI	14 (13.5)	10 (13.9)	10 (21.3)	15 (38.5)*#
Shock	7 (6.7)	4 (5.6)	5 (8.5)	5 (12.8)
IPN	17 (16.2)	8 (11.1)	9 (19.1)	8 (20.5)
Surgery	4 (3.8)	1 (1.4)	4 (8.5)	6 (15.4)*#
The length of ICU stay	9.8 ± 6.4	8.4 ± 5.6	12.2 ± 8.6#	13.6 ± 8.0*#
The length of hospital stay	14.6 ± 9.4	14.3 ± 8.8	17.2 ± 11.6	19.8 ± 13.5*#
Death	8 (7.7)	2 (2.8)	4 (8.5)	5 (12.8)

**P* < 0.05 compared with normal group; #*P* < 0.05 compared with mild HTG group; †*P* < 0.05 compared with moderate HTG group. ARDS, acute respiratory distress syndrome; AKI, acute kidney injury; IPN, infected pancreatic necrosis; ICU, intensive care unit.

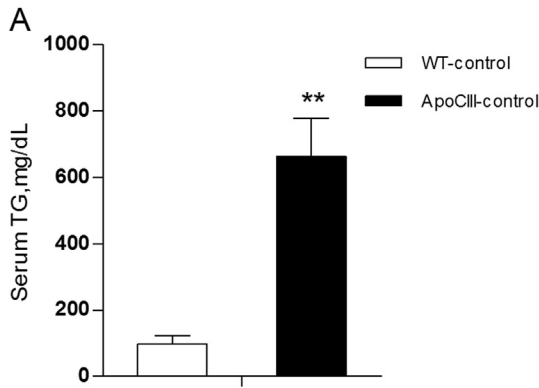


Fig. 1. Serum levels of TG between WT-control and ApoCIII-control groups. ** $P < 0.001$.

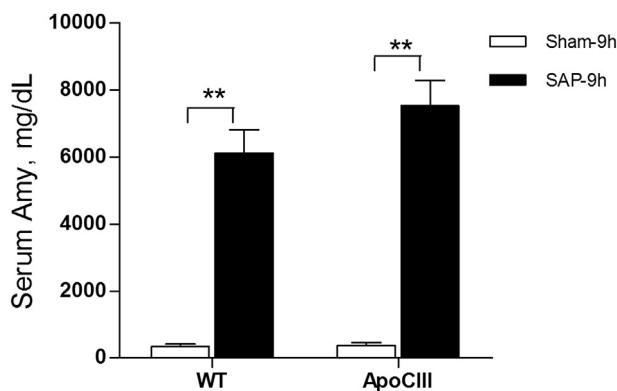


Fig. 2. Comparison of serum Amy between Sham-9h group and SAP-9h group. ** $P < 0.001$.

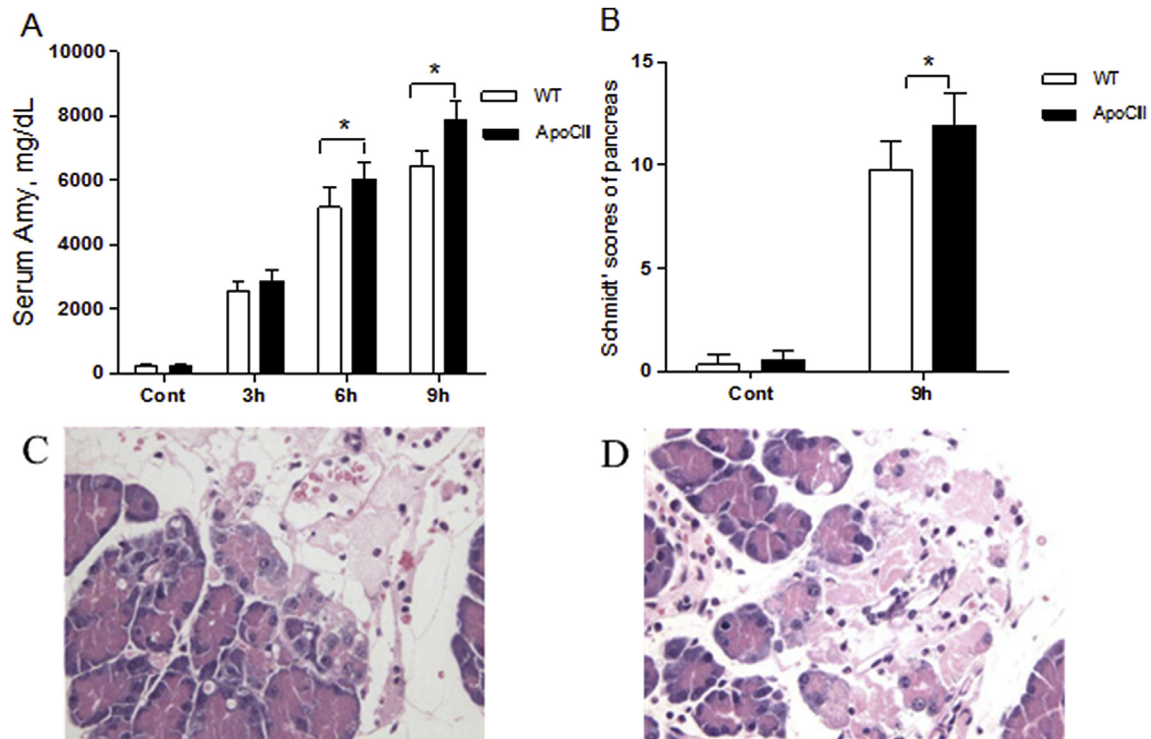


Fig. 3. Serum Amy (A) and histological scores (B) between ApoCIII-SAP groups and WT-SAP groups were compared. H&E staining of pancreas of WT-SAP-9h group (C) and ApoCIII-SAP-9h group (D) were observed under 400-fold magnification. * $P < 0.05$.

ApoCIII-SAP-6h group and ApoCIII-SAP-9h group were significantly higher than that of WT-SAP-6h group ($P < 0.05$) and WT-SAP-9h group ($P < 0.05$), respectively. The pancreatic histological score of ApoCIII-SAP-9h group was significantly higher than that of WT-SAP-9h group ($P < 0.05$) (Fig. 3B). Both groups had pancreatic edema and immune cells infiltrations, but acinar cells injury in ApoCIII-SAP-9h group was more serious (Fig. 3C and D).

ApoCIII-SAP groups had more serious kidney injury than WT-SAP groups

The serum levels of Scr (Fig. 4A) and BUN (Fig. 4B) presented increasing trends in both ApoCIII-SAP groups and WT-SAP groups. ApoCIII-SAP-9h group had significantly higher level of Scr than WT-SAP-9h group ($P < 0.001$); ApoCIII-SAP-6h group and ApoCIII-SAP-9h group had significantly higher level of BUN than WT-SAP-6h group ($P < 0.05$) and WT-SAP-9h group ($P < 0.001$), respectively. In WT-SAP-9h group (Fig. 5A), pathologic assessments under 400-fold magnification observed that the epithelium cells of tubule were edematous and the lumens of renal tubular were narrow. However, ApoCIII-SAP-9h group (Fig. 5B) had more lumens turning into dilatation, as the brush of lumen surface fell off in proximal convoluted tubules. The renal histological score of ApoCIII-SAP-9h group was significantly higher than that of WT-SAP-9h group ($P < 0.05$) (Fig. 5C).

Discussion

HTG has been regarded as a cause of AP since decades ago [3], the proportion of HTG in all the AP etiologies varies in different studies. Twenty years ago, a study by Fortson et al. [5] indicated that HTG was the etiology for 1.3–3.8% of the AP patients. A multi-center study displayed a decade ago in Taiwan showed that HTG-induced pancreatitis accounted for 12.3% of the AP episodes [22].

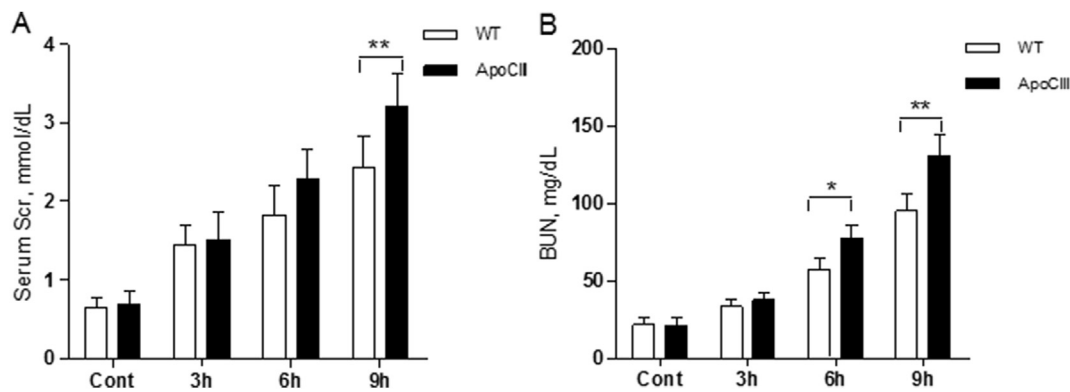


Fig. 4. Serum levels of Scr (A) and BUN (B) between WT and ApoCIII groups were compared. * $P < 0.05$, ** $P < 0.001$.

At the same time, another study involving 176 AP patients revealed that the proportion of the patients with admission serum TG above 500 mg/mL was 25.6% [23]. A latest study enrolling 2461 patients with AP in Beijing showed the proportion of HTG defined as serum TG more than 1000 mg/mL was 10.36% [24]. The prevalence of HTG in AP may be very different between ethnic groups. A recent clinical study from UK showed that patients with serum TG above 1000 mg/mL only take 2.3% in the population of AP [25]. Although there are different options about HTG as an epiphenomenon or a more common co-morbidity of AP [26], the general trend is that patients with high level of admission serum TG are increasing in China, and more attentions are attracted to hypertriglyceridemic pancreatitis.

In a study by Tariq et al. [27], patients with AP were divided into HTG group (TG ≥ 200 mg/dL) and control group (TG < 200 mg/dL); they found that a TG level of above 200 mg/dL on admission in AP is an independent predictor of developing complications, hospital LOS, admission to ICU, and ICU LOS. We further divided HTG into more groups, and our clinical study indicated that mild elevation in TG levels does not affect the severity of AP. Balachandra et al. [4] concluded HTG had no significant correlation with the clinical severity of AP. In their study, only 14 patients were defined as HTG of the 43 patients with AP, and the mean level of TG was only 265 mg/mL in HTG group. The outcome is consistent with our

result. Our previous study showed HTG is a risk factor for AKI in AP [15]. The clinical study of Hamada confirmed our result [28]. Actually, the incidence of AKI in patients with mild elevation of admission TG is not higher than that of normal group; only severe HTG can deteriorate renal function in the early phase of AP.

High-fat diet and intravenous administration of the detergent Triton-WR 1339 are common method to establish HTG models in previous studies [29–32]. Since mild HTG dose not aggravate disease severity of AP, a suitable animal model should be established. The following issues were taken into consideration according to the clinical characteristics: 1) the serum level of TG in experimental animals should be much higher than normal range; 2) edematous/mild pancreatitis models may be not tough enough to distinguish AKI (organ failure) between WT mice and hypertriglyceridemic mice. Although these mild or transient HTG models presented more severe pancreatitis than non-HTG animals, and potential mechanisms have been investigated and discussed, they are not appropriate to reflect clinical situations. Human ApoCIII^{tg} mouse is more suitable for animal experiments of hypertriglyceridemic pancreatitis. The overexpression of ApoCIII in transgenic mice results in moderate to severe HTG [33]. Organ failures in edematous pancreatitis models established by intra-abdominal injection of caerulein are often mild; it is hard to withstand the adverse role of HTG. By the same argument, too severe pancreatitis models are also

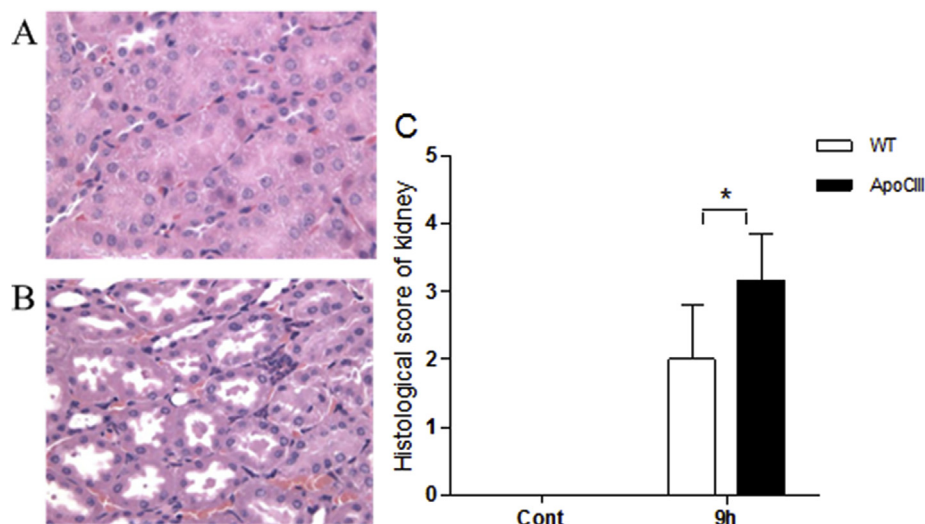


Fig. 5. H&E staining of kidney of WT-SAP-9h group (A) and ApoCIII-SAP-9h group (B) were observed under 400-fold magnification. The kidney histological score of ApoCIII-SAP-9h group was significantly higher than that of WT-SAP-9h group (C). * $P < 0.05$.

inappropriate. By exploring proper dose by retrograde injection of sodium taurocholate from duodenum to pancreatic duct, we determine to use 0.5% sodium taurocholate (0.1 ml/100 g weight) to induce model of pancreatitis.

In our animal experiment, ApoCIII^{tg} mice had more serious pancreatic damage and kidney injury than WT mice. Why severe HTG can exacerbate AKI in the course of AP has not been well elucidated. Lipotoxicity metabolized from HTG has been recognized as one potential cause of hypertriglyceridemic AP and pancreatic necrosis [3,34]. We previously supposed TG deposited in renal would be hydrolyzed by pancreatic enzyme in the early phase of AP, and lipotoxicity may accelerate the course of AKI [15]. The widely accepted pathophysiologic mechanisms of AP-related AKI involve inflammatory cytokines, intra-abdominal hypertension, microcirculation changes, and so on [35]. Our hypothesis is a supplement to these issues. The potential mechanisms involving dysregulation of lipid metabolism, lipotoxicity, and oxidative stress are expected to be investigated in hypertriglyceridemic AP-related AKI.

Although our clinical study did not get significant differences of ARDS, this result should be interpreted with caution. ARDS is one common complication during the early phase of SAP, and the incidence of ARDS increased with serum TG level in our study. The limited sample size of this study may be one reason for losing the expecting result.

In conclusion, mild HTG has little adverse impact on disease severity of AP; only severe HTG can aggravate kidney injury in the course of AP. ApoCIII-SAP mice have more serious pancreatic damage and kidney injury than WT-SAP mice. This animal model can be adopted for further investigations.

Disclosure

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